Recitation 2

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Outline

1. **Homework**

2. **Review**

3. **Recitation**

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Homework

1. A four-sided die is rolled repeatedly, until the first time (if ever) that an even number is obtained. What is the sample space for this experiment?

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The outcomes $\{a_1, a_2, a_3, \cdots, a_n\}$ where $a_1, a_2, \cdots, a_{n-1} \in \{1, 3\}$ and $a_n \in \{2, 4\}$ is enough? (infinite sequence is also possible!)

The sample space N^* (the number of rolling) is not the best.

Homework

3. Let $\{A_i : i \in I\}$ be a collection of sets. Prove the "De Morgan's Law":

$$\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c, \qquad \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c.$$

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Mathematical induction is OK, but you must prove the De Morgan's Law when $n = 2$. It is not obvious in this problem.

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Review

- Conditional probability

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

- Multiplication rule

$$\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \cdot \mathbf{P}(B \mid A) \cdot \mathbf{P}(C \mid A \cap B)$$

- Total probability theorem

$$\mathbf{P}(B) = \sum_{i=1}^{n} \mathbf{P}(A_i \cap B) = \sum_{i=1}^{n} \mathbf{P}(A_i)\mathbf{P}(B \mid A_i)$$

- Bayes' rule

$$\mathbf{P}(A_i \mid B) = \frac{\mathbf{P}(A_i)\mathbf{P}(B \mid A_i)}{\mathbf{P}(B)} = \frac{\mathbf{P}(A_i)\mathbf{P}(B \mid A_i)}{\sum_{j=1}^{n} \mathbf{P}(A_j)\mathbf{P}(B \mid A_j)}$$

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1. Homework

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3. **Recitation**

1. We roll two fair 6-sided dice. Each one of the 36 possible outcomes is assumed to be equally likely.

   (a) Find the probability that doubles are rolled.

   (b) Given that the roll results in a sum of 4 or less, find the conditional probability that doubles are rolled.

   (c) Find the probability that at least one die roll is a 6.

   (d) Given that the two dice land on different numbers, find the conditional probability that at least one die roll is a 6.

1. We roll two fair 6-sided dice. Each one of the 36 possible outcomes is assumed to be equally likely.

(*a*) Find the probability that doubles are rolled.

There are 6 possible outcomes that are doubles, so the probability of doubles is $6/36 = 1/6$.

1. We roll two fair 6-sided dice. Each one of the 36 possible outcomes is assumed to be equally likely.

   (b) Given that the roll results in a sum of 4 or less, find the conditional probability that doubles are rolled.

   The conditioning event (sum is 4 or less) consists of the 6 outcomes

$$\{(1,1),(1,2),(1,3),(2,1),(2,2),(3,1)\},$$

   2 of which are doubles, so the conditional probability of doubles is $2/6 = 1/3$.

1. We roll two fair 6-sided dice. Each one of the 36 possible outcomes is assumed to be equally likely.

(c) Find the probability that at least one die roll is a 6.

There are 11 possible outcomes with at least one 6, namely, $(6,6)$, $(6,i)$, and $(i,6)$, for $i = 1, 2, ..., 5$. Thus, the probability that at least one die is a 6 is 11/36.

Recitation

1. We roll two fair 6-sided dice. Each one of the 36 possible outcomes is assumed to be equally likely.

(d) Given that the two dice land on different numbers, find the conditional probability that at least one die roll is a 6.

There are 30 possible outcomes where the dice land on different numbers. Out of these, there are 10 outcomes in which at least one of the rolls is a 6. Thus, the desired conditional probability is $10/30 = 1/3$.

2. A coin is tossed twice. Is it right that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Does it make a difference if the coin is fair or unfair?

2. A coin is tossed twice. Is it right that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Does it make a difference if the coin is fair or unfair?

Let A be the event that the first toss is a head and let B be the event that the second toss is a head. We have

$$\mathbf{P}(A \cap B \mid A) = \frac{\mathbf{P}((A \cap B) \cap A)}{\mathbf{P}(A)} = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(A)}$$

$$\mathbf{P}(A \cap B \mid A \cup B) = \frac{\mathbf{P}((A \cap B) \cap (A \cup B))}{\mathbf{P}(A \cup B)} = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(A \cup B)}$$

Notice that $\mathbf{P}(A) \leq \mathbf{P}(A \cup B)$. Therefore,

$$\mathbf{P}(A \cap B \mid A) \geq \mathbf{P}(A \cap B \mid A \cup B)$$

3. You enter a chess tournament where your probability of winning a game is 0.3 against half of the players (call them type 1), 0.4 against a quarter of the players (call them type 2), and 0.5 against the remaining quarter of the players (call them type 3). You play a game against a randomly chosen opponent.

(a) What is the probability of winning?

(b) Suppose that you win. What is the probability that you had an opponent of type 1?

3. You enter a chess tournament where your probability of winning a game is 0.3 against half of the players (call them type 1), 0.4 against a quarter of the players (call them type 2), and 0.5 against the remaining quarter of the players (call them type 3). You play a game against a randomly chosen opponent.

(*a*) What is the probability of winning?

Let A_i be the event of playing with an opponent of type i and B be the event of winning. We have

$$\mathbf{P}(A_1) = 0.5, \quad \mathbf{P}(A_2) = 0.25, \quad \mathbf{P}(A_3) = 0.25.$$
$$\mathbf{P}(B \mid A_1) = 0.3, \quad \mathbf{P}(B \mid A_2) = 0.4, \quad \mathbf{P}(B \mid A_3) = 0.5.$$

By the total probability theorem, the probability of winning is

$$\begin{aligned}\mathbf{P}(B) &= \mathbf{P}(A_1)\mathbf{P}(B \mid A_1) + \mathbf{P}(A_2)\mathbf{P}(B \mid A_2) + \mathbf{P}(A_3)\mathbf{P}(B \mid A_3)\\ &= 0.375.\end{aligned}$$

3. You enter a chess tournament where your probability of winning a game is 0.3 against half of the players (call them type 1), 0.4 against a quarter of the players (call them type 2), and 0.5 against the remaining quarter of the players (call them type 3). You play a game against a randomly chosen opponent.

(*b*) Suppose that you win. What is the probability that you had an opponent of type 1?

$$\mathbf{P}(A_1|B) = \frac{\mathbf{P}(A_1)\mathbf{P}(B|A_1)}{\mathbf{P}(B)} = 0.4.$$

4. **The Monty Hall Problem.** You are told that a prize is equally likely to be found behind any one of three closed doors in front of you. You point to one of the doors. A friend opens for you one of the remaining two doors, after making sure that the prize is not behind it. At this point, you can stick to your initial choice, or switch to the other unopened door. You win the prize if it lies behind your final choice of a door. Consider the following strategies:

(a) Stick to your initial choice.

(b) Switch to the other unopened door.

(c) You first point to door 1, and your friend always chooses to open door 2 if the prize is behind door 1. If door 2 is opened, you do not switch. If door 3 is opened, you switch.

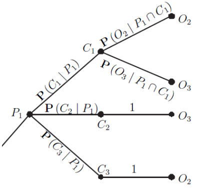
Which is the best strategy?

(a) Stick to your initial choice.

Your initial choice will determine whether you win or not, and the probability of winning is $1/3$. This is because the prize is equally likely to be behind each door.

(b) Switch to the other unopened door.

If the prize is behind the initially chosen door (probability $1/3$), you do not win. If it is not (probability $2/3$), and given that another door without a prize has been opened, you will get to the winning door once your switch. Thus, the probability of winning is now $2/3$.



(c) You first point to door 1, and your friend always chooses to open door 2 if the prize is behind door 1. If door 2 is opened, you do not switch. If door 3 is opened, you switch.

If the prize is behind door 1, you do not switch, and you win. If the prize is behind door 2, your friend opens door 3, you switch, and you win. If the prize is behind door 3, your friend opens door 2, you do not switch, and you lose. Thus, the probability of winning is 2/3.